Principal Component Analysis (PCA)

Feat → Feature Extract

↳ (i) Curse of dim
(ii) feature Selection
(iii) PCA Intuition
(iv) Geometrical Interpretation
(v) Scikit learn
(vi) Dataset Implementation.

Curse of dimensionality

4 column = dimension

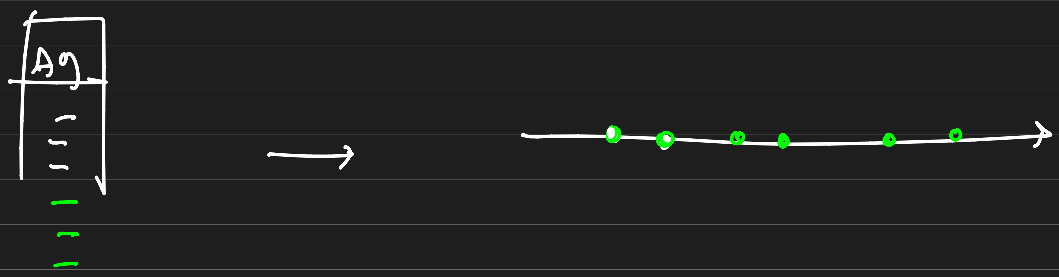
{feature}

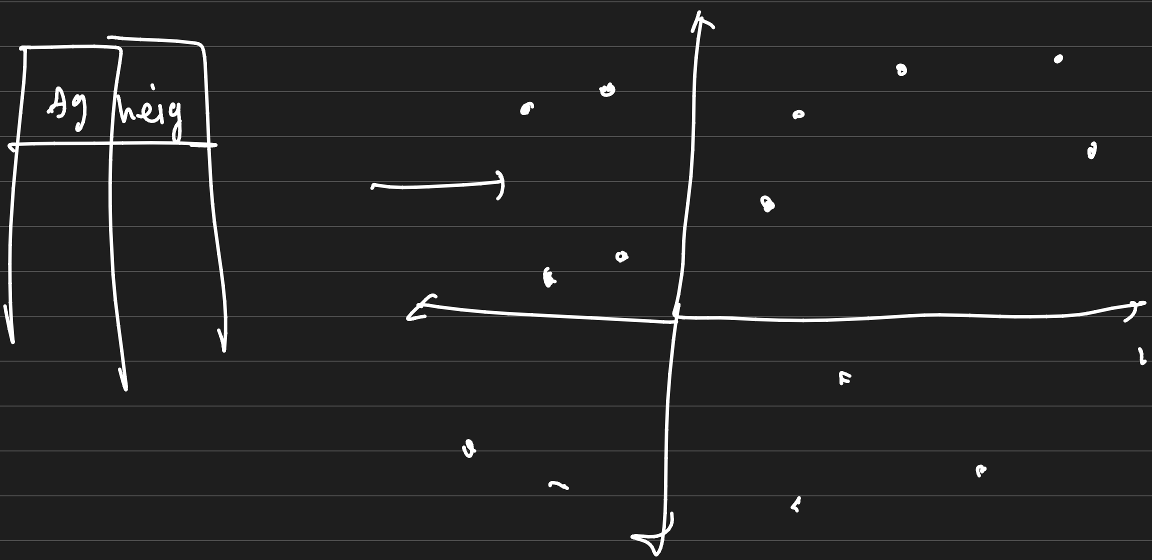
	Age	he	we	IQ
A →	10	3	4	
B →				
C →				

3 ta row

4 feat → 4 dimension.

5 feat → 5 dimension.



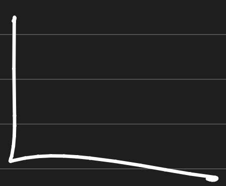


3D → , আরো বেশি spread করতে পারবে

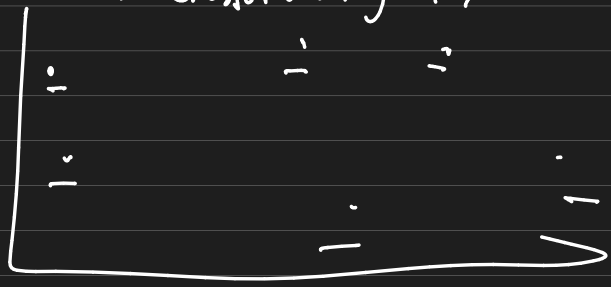
4D

5D

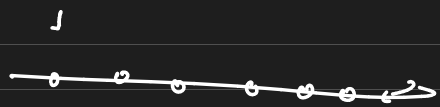
Data →

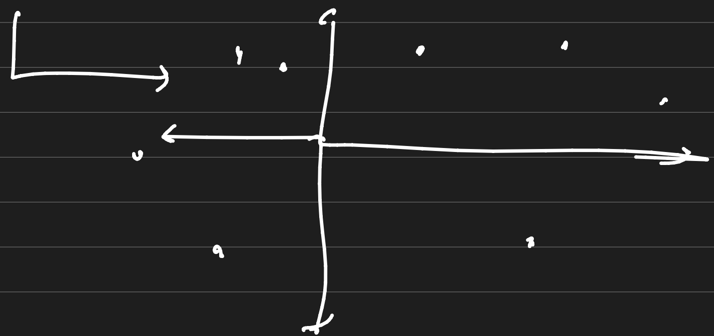


Dimensionality ↑↑ → Data spread ↑↑

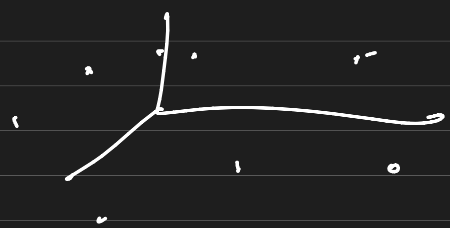


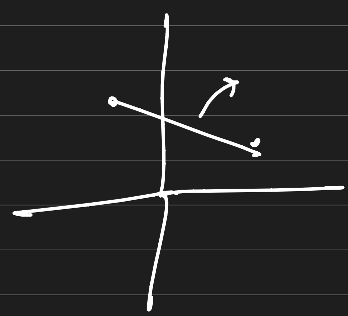
100 →





100 →





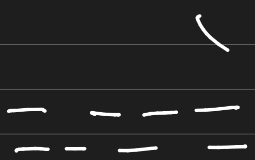
Distance →

feature → PCA

Principal Component Analysis (PCA)

Data এর তুলনায় feat ↑↑

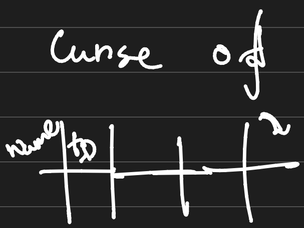
Dimensionality → curse of dimensionality
- ↳ (i) Distance based (algo)
- (ii) → Linear Reg → $\underline{w}^r$
- (iii) Tree based ↑



→ (ii) higher memory usage

↳ (iii) Slower training.

Curse of dimensionality → ① data points ↑↑
- (ii) feature Selection (Manually)
- (iii) feature Extraction (Auto)
  - ↳ PCA

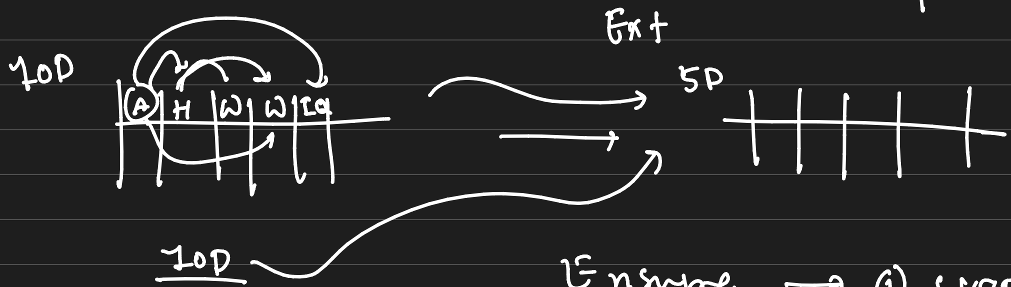


PCA (Principal Component Analysis) ↗ Unsupervised

Principal ↓ মৌলিক

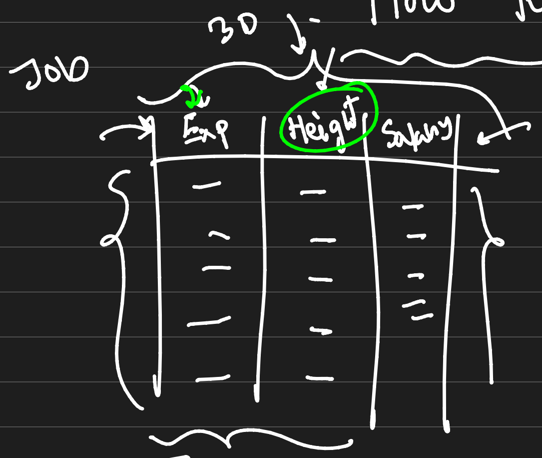
↳ principle feature Extraction.

↳ Given feat $\xrightarrow[\text{Ext}]{\text{feat}}$ Principal feat.

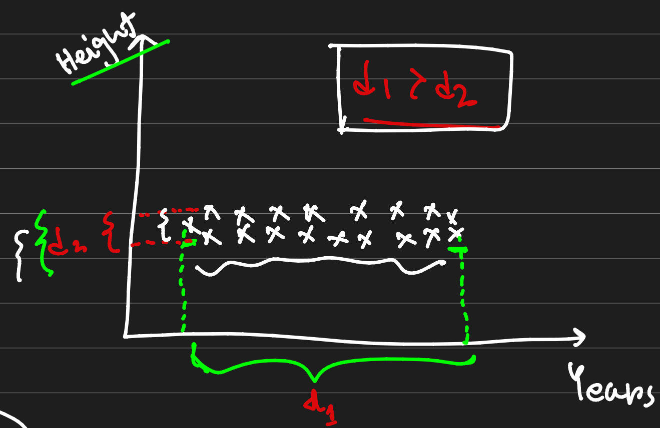


Ensure → ① সবচেয়ে কম data loss হবে

How to select features (Basic Way)

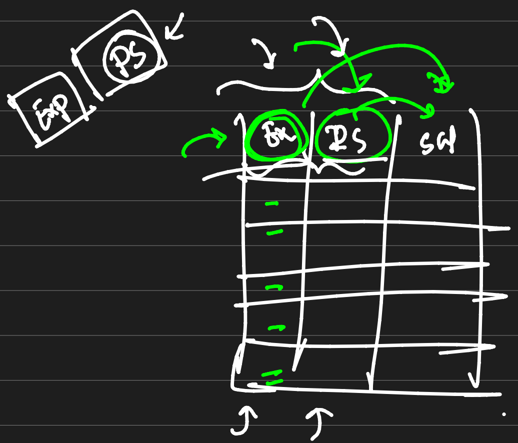


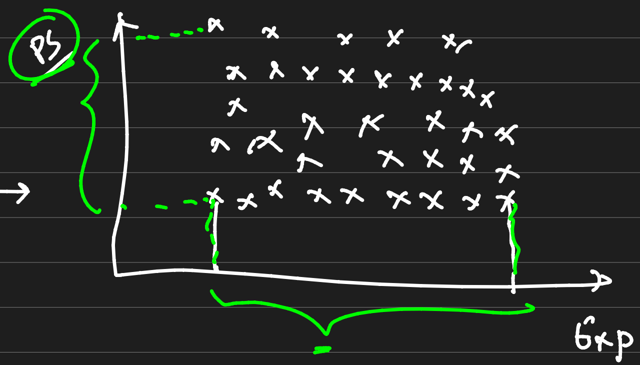
2D → 1D



$d_1 > d_2$

Choose → Exp

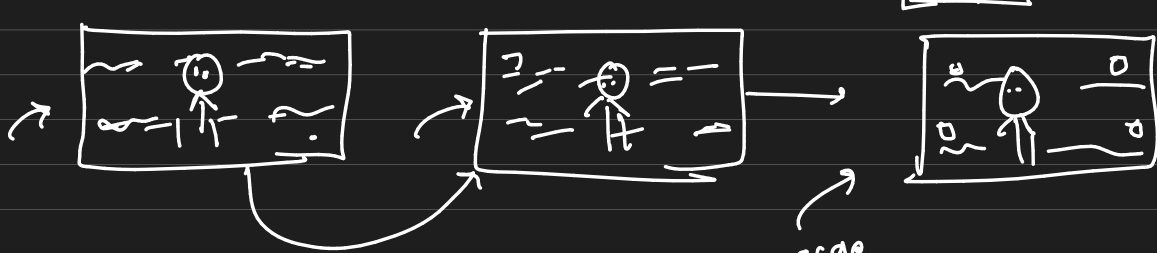




Concept → PCA → PC_1 PC_2

Data loss ↓

Camera → 4K → 1080p → 720p

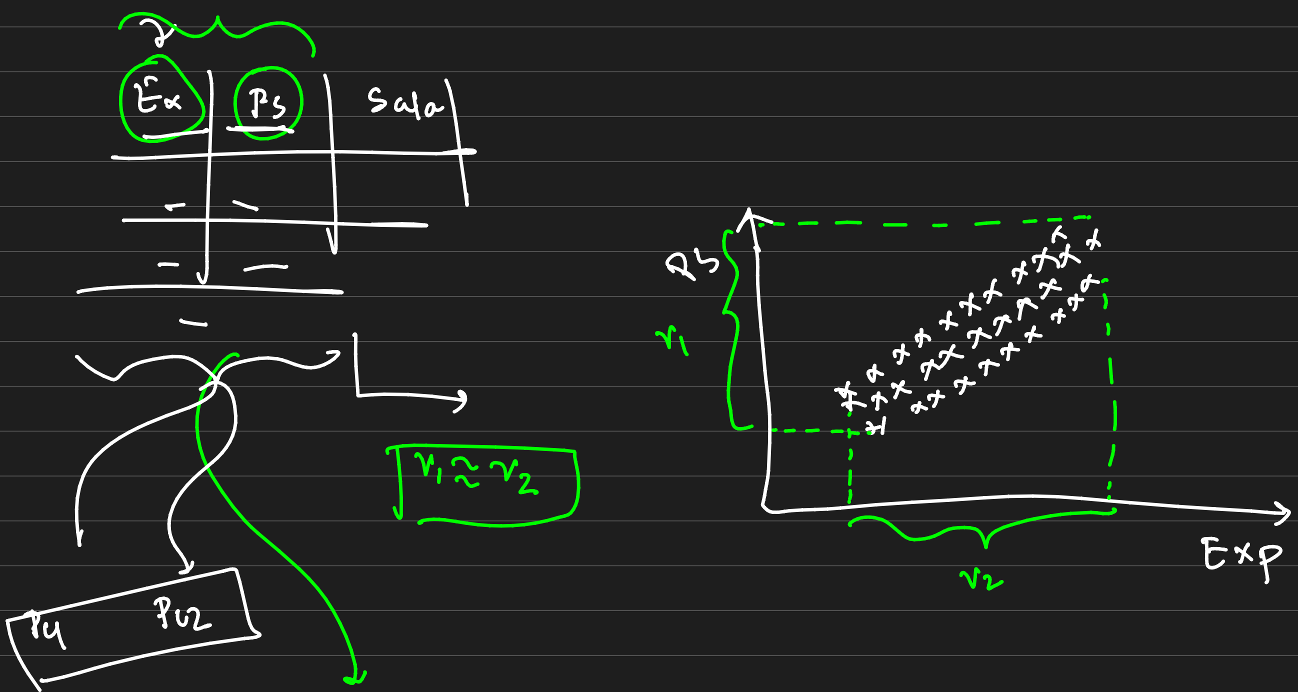


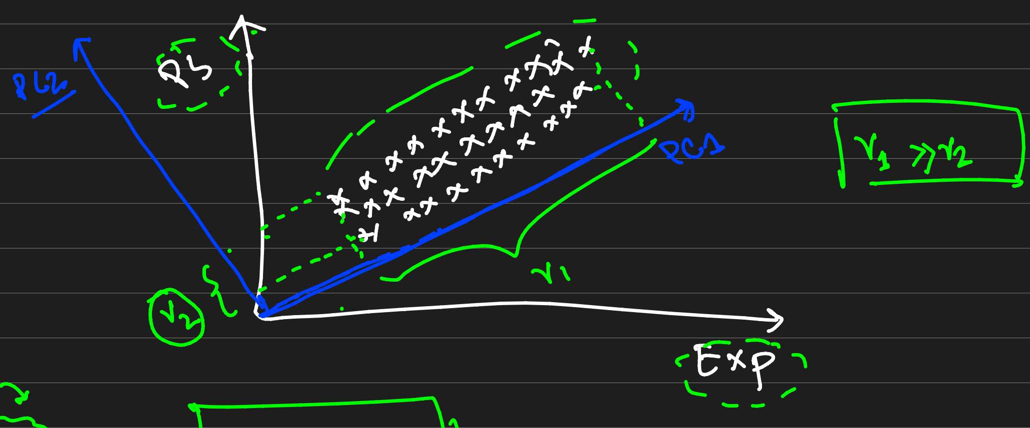
Message

Working of PCA

① Variance is important.

↳ PCA makes n {Principal components} that holds max variance.





PC1 → 90% Var

PC2 → 10% Var

Data var

(ii) Makes n' number PC combining correlated features

↳ Ensuring data's essence.

(iii) Transforms existing data based on Principal component.

PC1	PC2	Sal

$\rightarrow$

PC1	Sal

90% var

feature extraction.

feat $\rightarrow$ 10
PC $\rightarrow$ 10

$\rightarrow$ Age, height, weight, IQ . . . . .

$\rightarrow$ PC1 PC2 PC3 PC4 . . . PC10

PC

Age, height, weight

height, weight $\rightarrow$ PC2

Age	height	Weight	PC1

Data $\rightarrow$

PC1	PC2	PC3	Pc .

– –

PC10

90%

PC1 $\rightarrow$ 70% var
PC2 $\rightarrow$ 20% var
$\rightarrow$ 90% var

PC3 $\rightarrow$ 7%
PC4 $\rightarrow$ 5%
⋮
PC10

10%

10 feat
$\downarrow$
2 feat

Memory
speed
fix COD

Math → Covariance Mat, Eigen values, Eigen vector

Transformation of Matrix

PCA ভালো কাজ করে না → (i) **correlation** কম থাকলে

→ (ii) $\boxed{\text{Acc} \downarrow\downarrow}$

Pred effect আছে এই ধরনের feat delete করলে

Let's code PCA

Var

Var ↓

PC_1

PC_2

PC_3

PC_4

PC_5

⋮

PC_{31}